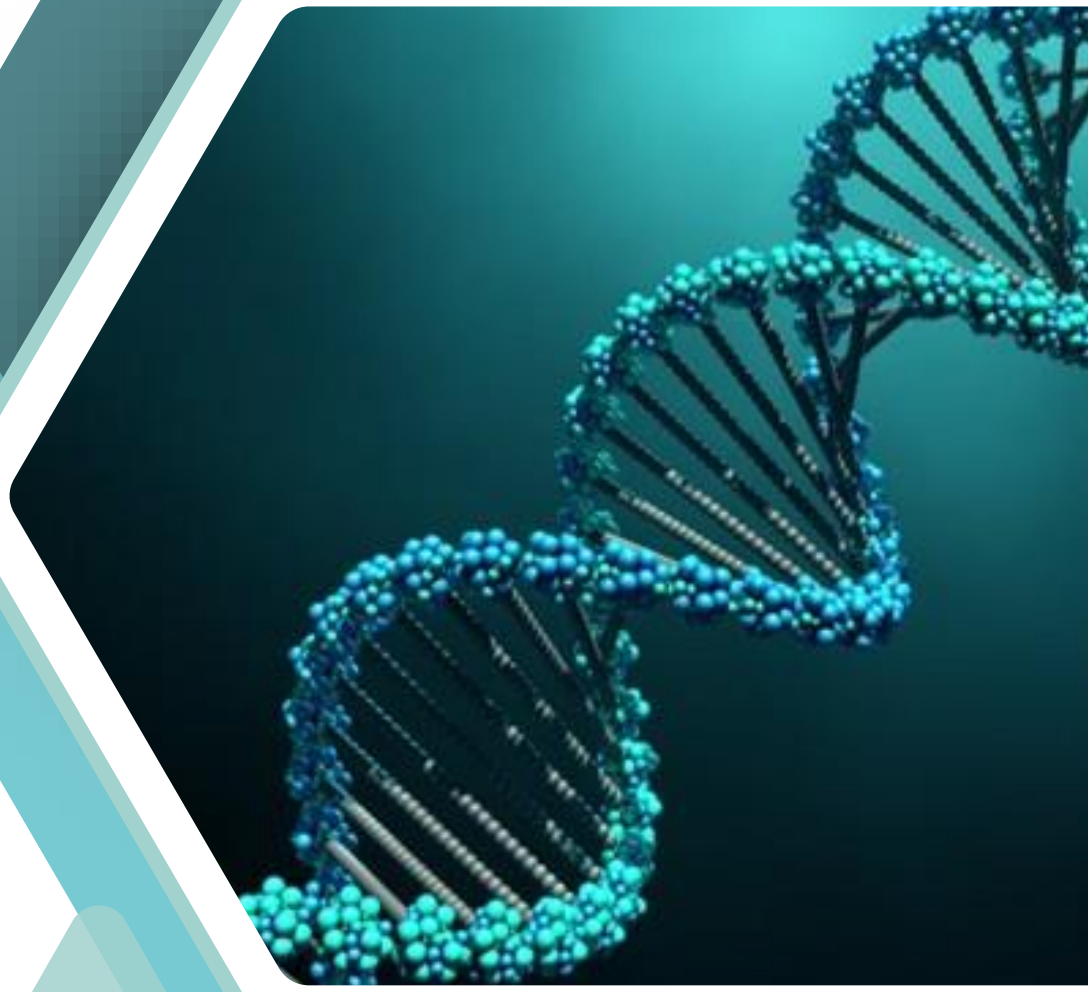
OCTAVUS CONSULTING
Data.Insights.Decisions.

GENE THERAPY HIGHLIGHTS Within the CGT Insights Corner 2025-26



Month: May 2026
Presenter: Team Octavus



Content

- Scientific Foundations of Gene Therapy 05
- Market Pulse of Gene Therapy 10
- Gene Therapy: Landscape & Upcoming Milestones 15
- Regulatory & Approval Landscape 20
- Tracking Spark Therapeutics Growth Within Gene Therapy Ecosystem 23
- Challenges, Risks & Strategic Assessment 25
- Future Frontier of Gene Therapy 28
- Glossary of Abbreviations 30

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

PRIOR ISSUES

- Highlights from AACR-25 (May, 25)
- US Tariff impact on Pharma industries (Jun, 25)
- EULAR-25: Key insights of Psoriatic Arthritis (Jul, 25)
- Antibody Drug Conjugate: The next big investment in Oncology (Aug, 25)
- Animal Health (Oct, 25)
- Obesity deep dive (Nov ,25)
- GLP-1 and GIP (Dec, 25)
- Hematological Highlights from ASH 2025 (Jan, 26)
- China Biotech's Global Positioning from JPM-26 (Feb,26)
- BIOSIMILARS RISING: The Catalyst For A New Age In Biologics Therapies (Mar,26)
- Cell Therapy Within CGT **(Apr, 26)**

Additional: 1 Beautiful Bill

INSIGHTS CORNER

Executive Summary

Part 2 of the CGT landscape, with primary focus on gene therapy

From experiment to platform: Gene therapy has shifted from high-risk experimentation to a strategic platform across rare diseases, hemoglobinopathies, ophthalmology, neuromuscular and emerging CNS indications

Diverging technology archetypes and Vectors as core infrastructure: The landscape is fragmenting across *in vivo* vs *ex vivo*, AAV vs non-viral/LVV, and gene-editing modalities (CRISPR, base editing), creating distinct competitive archetypes and vectors sit at the center of the biotech stack with AAV being the workhorse for durable *in vivo* delivery

Capital and deal momentum: The funding and deal environment is maturing, with ~$24B in disclosed funding across 154 rounds and ~$25B in 2025 deal value, skewed toward late-stage and approved assets in oncology and rare disease platforms

Execution as the bottleneck: Vector supply, CMC comparability, immunogenicity, redosing constraints and manufacturing scale-up are now as decisive as the headline clinical data in determining long-term winners

Clinical and regulatory tailwinds: Gene therapy trials are rising steadily (over 160 new trials annually by 2025) in major markets, while FDA, EMA and ICH roll out flexible CMC, rare-population trial and long-term follow-up frameworks that de-risk development





Scientific Foundations of Gene Therapy

(Core concepts, technologies & therapeutic strategies)

OCTAVUS CONSULTING
Data.Insights.Decisions.

INSIGHTS CORNER

Gene Therapy: Key Approaches

Decoding the scientific approaches behind advanced genetic therapies

Gene therapy is an advanced therapeutic approach to deliver exogenous genes that compensate for or replace the function of defective genes., offering a potential treatment strategy beyond conventional therapies such as pharmaceuticals, radiotherapy or surgery

Gene Addition/ Gene Augmentation

Gene addition is a widely explored gene therapy approach for treating monogenic disorders caused by mutations in one or both copies of a gene. The strategy involves delivering functional or healthy copies of the gene into a patient's cells using a vector.

E.g. Hemophilia B is treated by delivering a functional copy of the clotting factor IX (FIX) gene into liver cells using viral vectors, with the goal of restoring normal blood clotting activity.

Gene Editing

Gene editing is a technique of creating specific breaks in the DNA using specialized techniques, with or without repair signals. Two main approaches to gene editing are gene disruption/inactivation (gene is switched off) & gene correction/insertion (gene is repaired or a new genetic sequence is added).

E.g. CRISPR-Cas9 has become most widely adopted gene-editing technology due to its high precision & efficiency.

Gene Silencing/ Gene Inactivation

Gene silencing or gene knockdown, is an approach used to reduce or inhibit the expression of specific disease-causing genes at the transcriptional or translational level.

Antisense technology & RNA interference (RNAi) are most widely used strategies for achieving gene knockdown.

E.g. In Huntington's disease, RNAi can target the mutant huntingtin gene by degrading its mRNA, thereby lowering the production of the toxic huntingtin protein.

Gene Replacement/ Gene Correction

Gene replacement is a therapeutic approach that restores normal cell function by replacing a defective gene with a functional copy. This strategy is particularly useful for treating genetic disorders caused by mutations that result in the absence or malfunction of specific proteins.

E.g. Treatment of severe SCID, commonly referred to as "bubble boy disease".

INSIGHTS CORNER

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

Types of Gene Therapy

Somatic Gene Therapy

Somatic gene therapy corrects genetic disorders via targeted modification of non-germline somatic cells. The approach typically involves *ex vivo* isolation of dysfunctional cells, followed by their transduction or transfection with a functional wild-type gene copy. These genetically corrected cells are then reinfused into the patient, where they engraft and restore the deficient gene product's function

Key Features

- Non-heritable: Changes stay within the treated patient only
- Minimal ethics: Genetically straightforward, widely accepted
- Clinically proven multiple US FDA/EMA: Approved treatments exist

Approved Molecules

KRELSADI, ITVISAM, WASKYRA, ZOLGENSMA, LUXTURNA, KYMRIAH, YESCARTA, HEMGENIX, SKYSONA are some of the approved therapies

Germline Gene Therapy

Germline gene therapy targets reproductive cells (sperm, eggs, or early embryos) to permanently alter heritable genetic makeup, aiming to prevent disease or introduce enhancing alleles. Unlike somatic approaches, it remains highly controversial, with no human trials conducted due to an informal moratorium in the scientific community

Key Features

- Heritable: Changes pass to all offspring & descendants
- Unpredictable: Long-lasting, irreversible, species-wide effects
- Global ban: Not in experimental stage; ethically prohibited

Approved Molecules

No jurisdiction has approved germline genome editing due to safety & ethical concerns. Modifications to germline cells become heritable, potentially perpetuating adverse effects across generations. Hence, most EU nations have ratified the Oviedo Convention, banning heritable human genetic modification

INSIGHTS CORNER

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

Ex Vivo to *In Vivo*: Redefining the Future

Why Shift is happening:

Scientific Advances: Better delivery systems with advances in viral vectors (AAV, lentivirus) and non-viral delivery (LNPs, nanoparticles). CRISPR & base-editing tools optimized for *in vivo* use.

Economic Pressure: Reduced manufacturing complexity & cost, with greater scalability to treat larger patient populations.

Patient Needs: *In vivo* approaches offer the potential for faster treatment, reduced procedural invasiveness & broader access for patients.

Strategic Implications:

- *In vivo* gene therapy is poised to rapidly capture monogenic disease indications e.g., Hemophilia, Metabolic disorders & Ophthalmic conditions given the established clinical track record of AAV-based delivery in these settings
- However, *Ex vivo* autologous approaches will remain the dominant modality in hematologic oncology in the near term, as CAR-T efficacy data are too strong to displace without equivalent *in vivo* data

MODALITY COMPARISON

Dimension	*Ex vivo*	*In vivo*
Targeting precision	High-defined cell population	Variable-off-target tissues a concern
Immune evasion	Favourable, avoids allogenic rejection	Challenging; pre-existing AAV antibodies limits eligibility
Manufacturing	Patient-specific, vein-to-vein 30-90 days	Off-the-shelf vector production; weeks to manufacture at scale
Cost of Goods	Very high	Moderate
Scalability	Low	High
Approved products	CAR-T (KYMRIAH, YESCARTA), gene-modified HSCs (ZYNTEGLO, CASGEVY)	AAV (LUXTURNA, ZOLGENSMA, ROCTAVIAN, HEMGENIX); LNP-mRNA (CASGEVY-adjacent)
Therapeutic areas	Oncology (haem), monogenic blood disorders, autoimmune (emerging)	Ophthalmology, neurology, hepatology, rare metabolic; expanding to CNS, muscle
Market access	Restricted, needs specialized treatment centers and cold-chain logistics	Broader, simpler logistics, outpatient potential for some indications
Redosability	Autologous re-manufacturing costly	Limited; anti-capsid antibodies typically prevent redosing with same serotype

INSIGHTS CORNER

From Viruses to Nanocarriers

Viral vs Non-viral Vectors

Vectors serve as a delivery backbone of CGT, enables transfer of genetic material to modify or correct defective genes for the treatment of genetic disorders, cancers & other diseases. In the 21st century, advances in viral vector manufacturing have significantly improved the safety & efficiency of these therapies. Today, both viral & non-viral vectors continue to drive innovation & expand the therapeutic potential of gene therapy

Viral Vector

Viral vectors offer highly efficient & targeted gene delivery, enable long-lasting gene expression and are generally non-replicating & minimally destructive to cells. However, their limited loading capacity restricts the delivery of larger genes & may induce immunogenic responses.

At present, viral vectors are the most commonly used delivery vehicles in US FDA approved therapies.



Non-viral Vector



Non-viral vectors offer advantages such as low toxicity, reduced mutagenesis risk, cost-effective, scalable production and low immunogenicity, providing key safety benefits over viral approaches. However, they often exhibit lower transfection efficiency, limited target specificity & delayed gene expression timelines.

Most widely researched non-viral delivery methods include chemical disruption, electroporation & polymer-based vectors.

Comparison of viral and non-viral vectors

Parameters	Viral	Non-viral
Immunogenicity	Highly immunogenic	Less immunogenic
Tissue transduction	Easily transduce different cells & tissues	Poor transduction potentials
Genetic modification	Requires genetic mutations	May not require genetic alteration
Chemical modification	Tentative	Required
Cell type	Somatic & germline	Mostly somatic
Examples	Retroviral, lentiviral & adeno-associated viral vectors, etc.	LNPs, Plasmid, virosomes, etc.

INSIGHTS CORNER

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com



Market Pulse of Gene Therapy

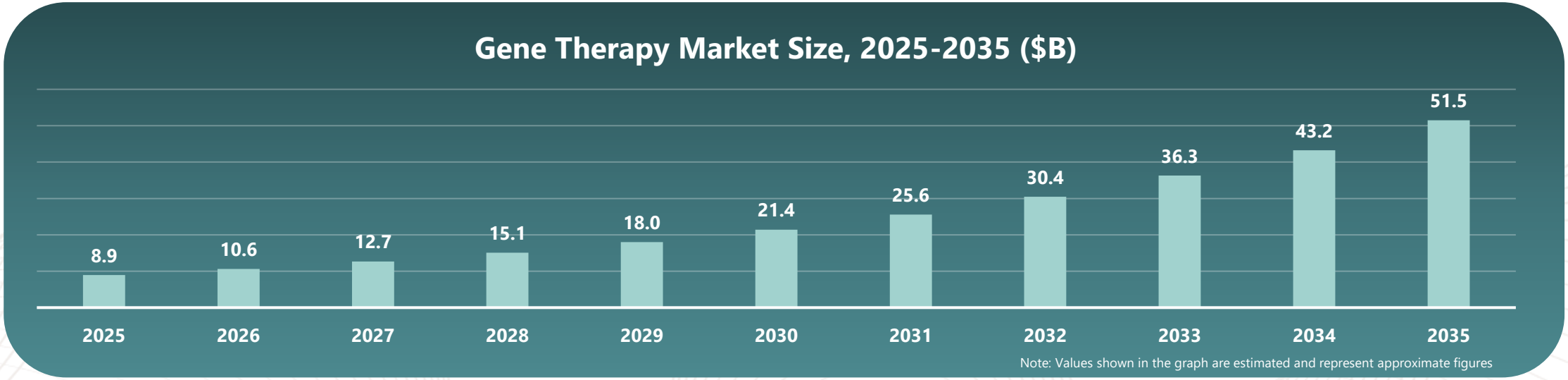
(Market landscape, growth drivers, investments & commercial trends)

OCTAVUS CONSULTING
Data.Insights.Decisions.

INSIGHTS CORNER

Gene Therapy: Market at a Glance

Evaluating the growth trajectory of rapidly evolving gene therapy landscape



KEY HIGHLIGHTS

- The gene therapy market is skyrocketing, currently worth up to $8.9B, it is expected to hit $51.5B by 2035, growing at roughly 19% CAGR annually, it's one of the fastest-growing sectors in all of healthcare market
- Use of harmless viruses (viral vectors) is the most common & successful delivery method in the market to carry healthy genes into cells. While these are the current standard, newer tools like CRISPR are making these deliveries much more precise & effective

KEY INSIGHTS

- **Curing vs Managing:** The market is exploding because medicine is shifting from just managing symptoms to actually fixing the genetic root cause of diseases
- **Innovation vs Cost:** While the growth potential is massive, the multi million-dollar price tag per treatment is the biggest hurdle standing in the way of these cures reaching most people

INSIGHTS CORNER

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

Why Gene Therapy Vectors are at the Centre of Global Biotech Revolution?

Gene therapy vectors sit at the centre of today's biotech revolution because they have become the critical "infrastructure" that enables curative, one-time genetic medicines at scale across multiple diseases & geographies

From niche experiments to a multi-billion dollar modality

- Gene therapy vectors have evolved from experimental tools to scalable platforms, with 2,000+ programs across major diseases
- Viral & non-viral vectors are now core biopharma growth drivers, with the global market projected to roughly 3x by 2030

Viral vectors as workhorse platforms (especially AAV)

- Viral vectors, particularly AAV, are central to gene therapy due to their reliable & long-term gene delivery capability
- AAV underpins many approved and late-stage therapies and is the default platform choice for rare genetic diseases

Non-viral and LNP systems moving beyond vaccines

- Non-viral vectors (e.g. LNPs, electroporation) are transforming gene therapy into scalable, repeat-dose treatments beyond niche use
- This is validated by LNP-enabled siRNA therapy (ONPATTRO) and mRNA vaccines, demonstrating rapid manufacturing and large-scale clinical impact

Convergence with CRISPR and *in vivo* Gene editing

- Gene therapy vectors enable CRISPR's clinical use by ensuring targeted delivery, controlled activity, and manageable immune response
- Advances in viral (AAV, adenoviral) & non-viral (LNPs) platforms are transforming CRISPR into a scalable, repeat-dose, systemic therapeutic modality

INSIGHTS CORNER

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

Funding Dashboard 2025

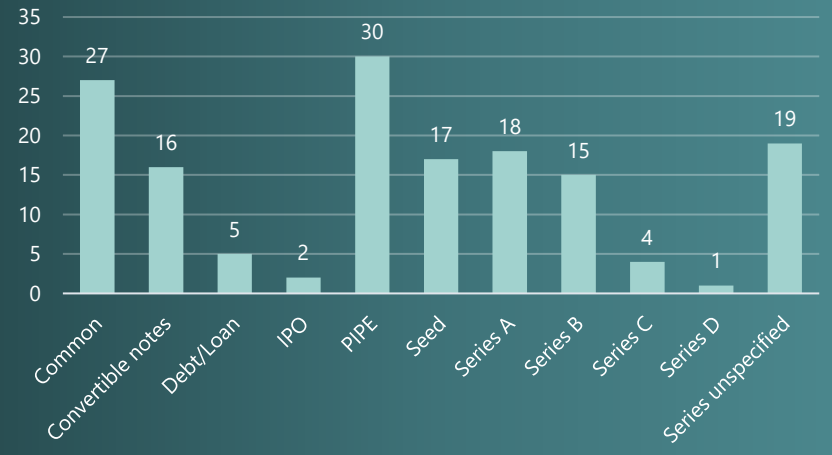
Key Highlights

- Total $24B disclosed amount across 154 fundings
- The largest funding round raised by Eli Lilly was $1.25B
- Out of 154 fundings, 142 are disclosed & 12 are undisclosed
- Oncology leads all therapeutic areas, with 77 deals and a total of $17.8B dollars in funding

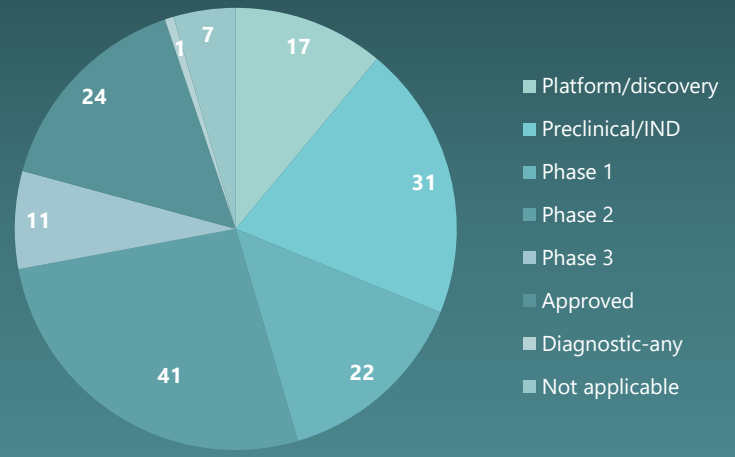
Key Insights

- Eli Lilly raised the largest round at $1.25B, showing dominance of big pharma
- Gene therapy is transitioning from research to the commercialization phase
- Phase 2 and preclinical stages show the strongest funding momentum within the clinical-stage pipeline
- Approved-stage therapies attract the highest funding, reflecting investor preference for lower risk
- There is a shift toward late-stage investments over early-stage research

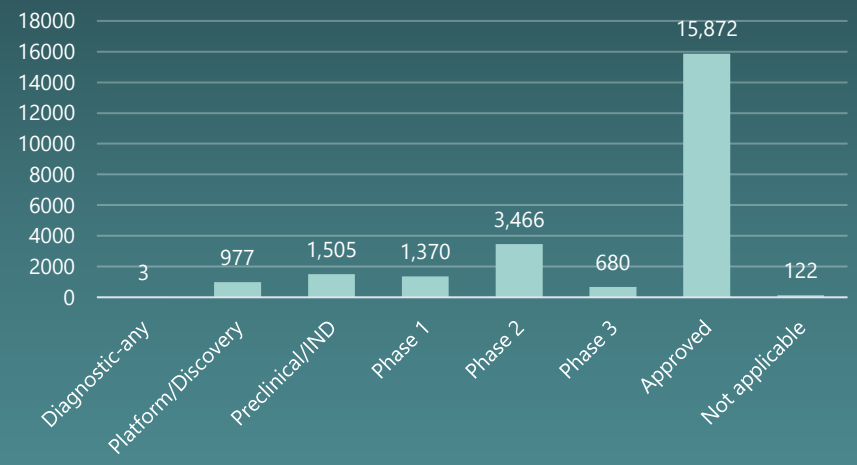
Funding Count by Round Type



Funding Count By Development Stage



Funding Amount By Development Stage ($M)



Top 5 Biotech Companies By Funding

Sl. No.	Company	Amount	Round	Stage
1	Eli Lilly and Co.	$1,250M	Convertible Notes	Approved
2	Ionis Pharmaceuticals	$700M	Convertible Notes	Approved
3	Sarepta Therapeutics	$600M	Debt/Loan	Approved
4	BridgeBio Pharma	$500M	Convertible Notes	Approved
5	Beam Therapeutics	$500M	Common	P2

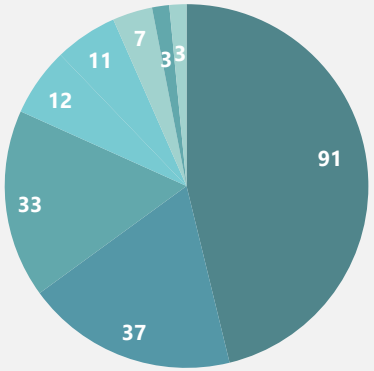
INSIGHTS CORNER

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

Deal Dashboard 2025

Total Deal Number	Total Deal Value	Total Milestone	Rare Disease	Largest Deal
197	$25.3B	$14.9B	29%	$5.6B
Across all deals	Aggregate $B	Per disclosed deal	% across all GT deals	SL Bio / Horizon Space

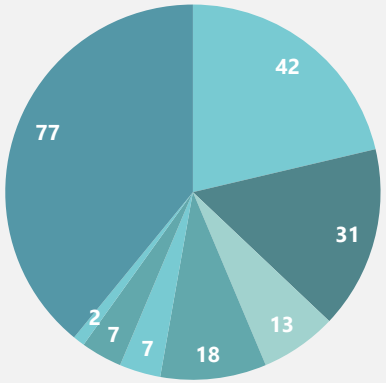
MAJOR DEAL TYPE BREAKDOWN



- Contract service
- University/foundation
- Dev & comm license
- Acquisition
- Others
- R&D only
- Company formation
- Platform, product purchase

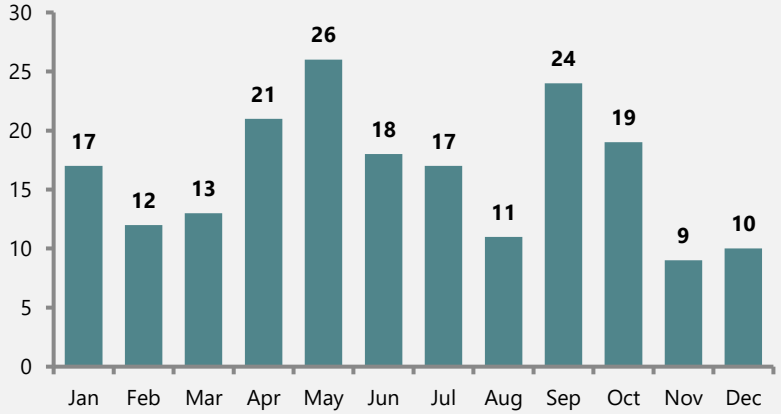
Others contains: deals with 1 & 2 number

STAGE AT DEAL SIGNING

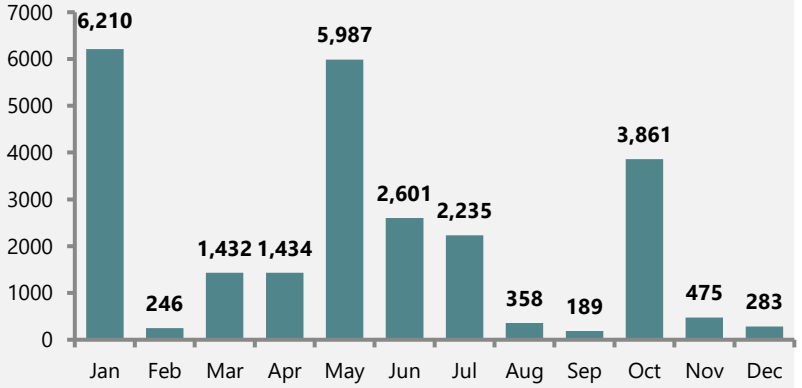


- Platform/discovery
- Preclinical/IND
- Phase 1
- Phase 2
- Phase 3
- Approved
- Diagnostic - any
- Not applicable

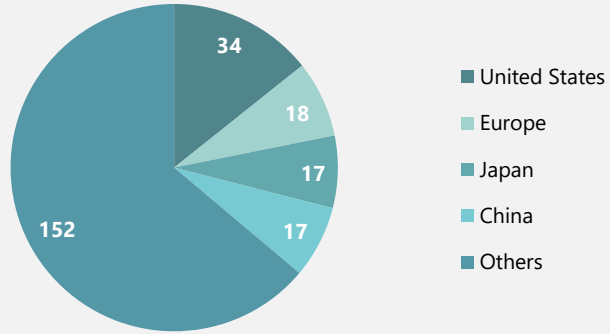
MONTHLY DEAL NUMBER (2025)



MONTHLY DEAL VOLUME ($M)



DEALS BY MAJOR MARKETS



Other deal contains: Not Applicable (132) & Undisclosed (22)
Note: Please note that a single deal may span multiple geographies; however, the figures presented reflect only the exclusive count for each geography

TOP 5 DEALS BY VALUE($M)

1	**SL Bio / Horizon Space** Oncology	$5,568
2	**Orna / Vertex Pharma** Hematology	$4,350
3	**Arbor Bio / Chiesi** Hepatology	$2,115
4	**Sangamo / Eli Lilly** Neurology	$1,418
5	**Keyi Pharma / Erigen** Oncology & Hematology	$1,335

INSIGHTS CORNER

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com



Gene Therapy: Landscape & Upcoming Milestones

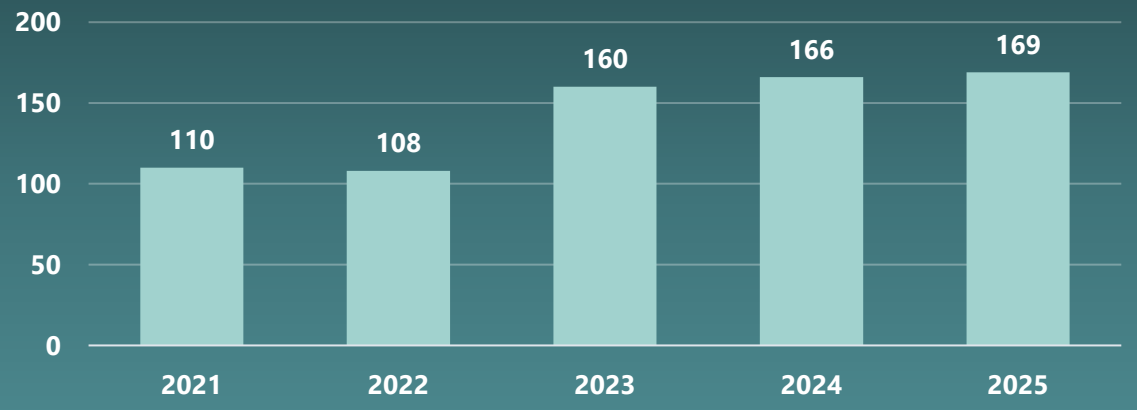
(Highlighting the evolving clinical pipeline & anticipated future milestones in gene therapy)

OCTAVUS CONSULTING
Data.Insights.Decisions.

INSIGHTS CORNER

Gene Therapy: Clinical Trials Overview

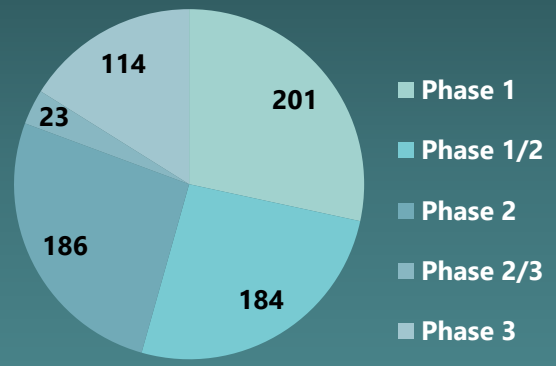
Number of Trials by Year



Number of Trials by Status



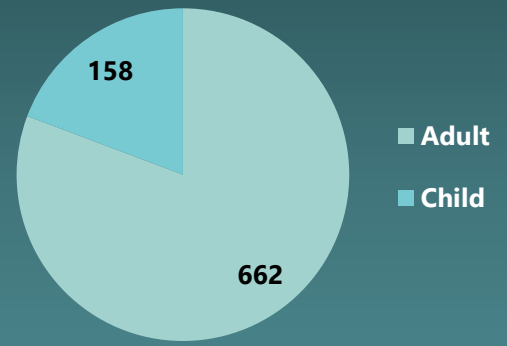
Number of Trials by Phase



Number of Trials by Region



Number of Trials by Population

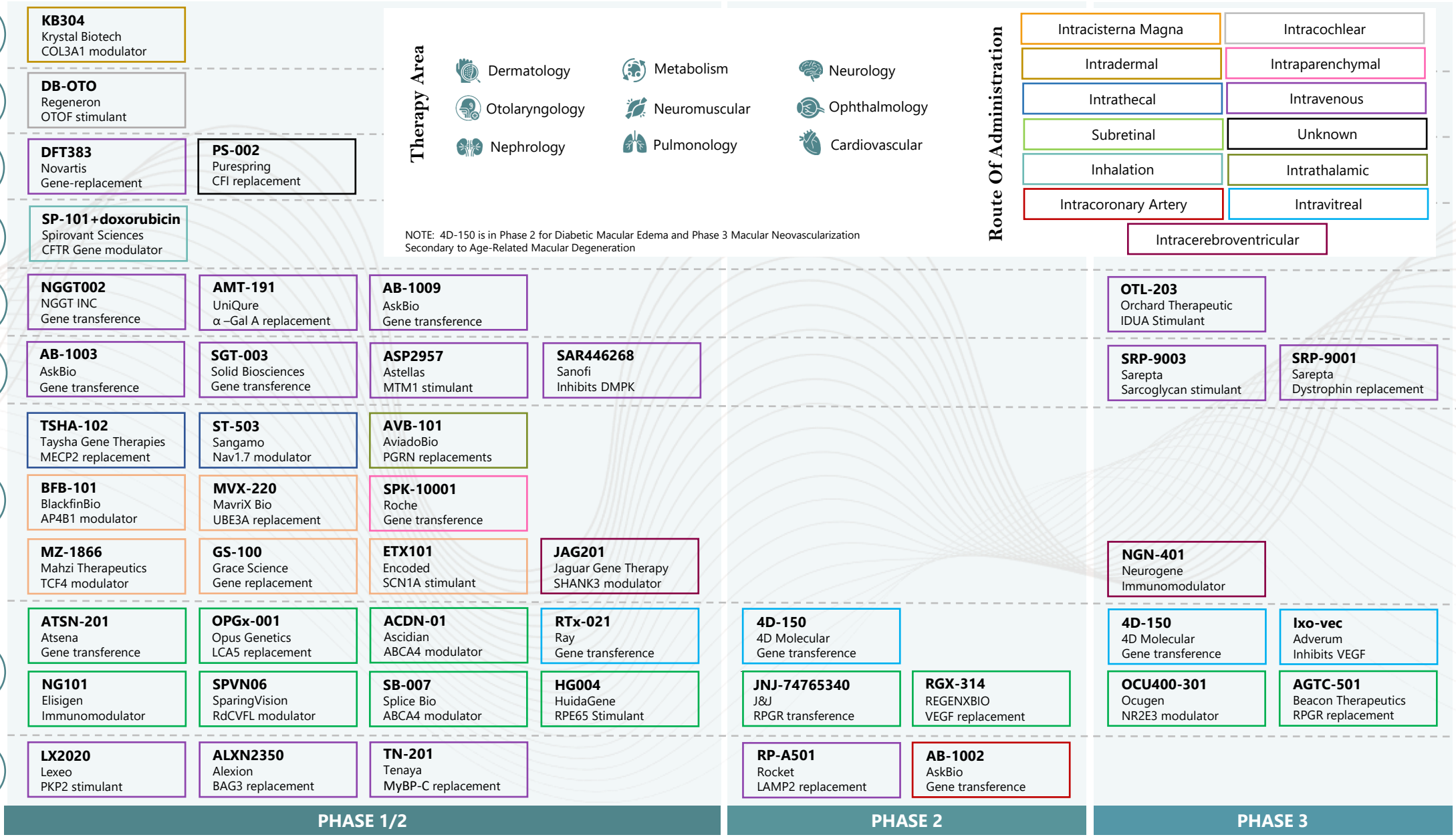


Note: Data has been collected for 5 years (Jan 2021 – Dec 2025) and covers interventional industry studies; EU 5 includes France, Spain, Germany, Italy and UK; Asia 4 includes Japan, China, South Korea, India

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

INSIGHTS CORNER

US Clinical Landscape (April 2023 - March 2026)



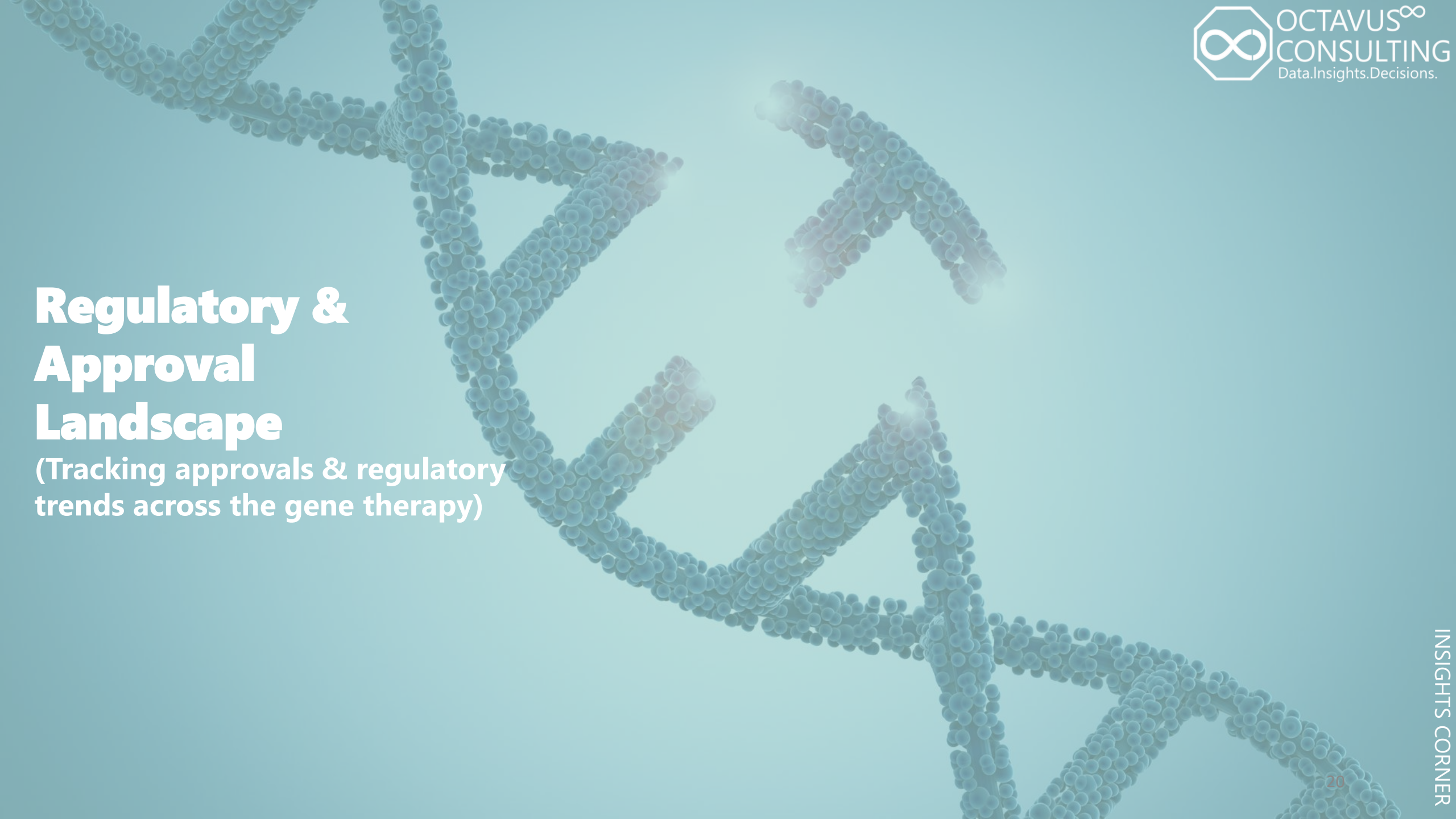
INSIGHTS CORNER

Gene Therapy 2026: What's on the horizon? (1/2)

Asset	Company	Indication	Upcoming Milestones
CTX310	CRISPR THERAPEUTICS	sHTG & Refractory hypercholesterolemia	Expected to provide a Phase 1b clinical update in H2'2026
CTX611 (SRSD107)	CRISPR THERAPEUTICS	TKA	Phase 2 clinical update anticipated in H2'2026
CTX460	CRISPR THERAPEUTICS	AATD	Expected to initiate a clinical trial for CTX460 in mid-2026
CTX340	CRISPR THERAPEUTICS	Refractory hypertension	Clinical trial initiation anticipated in H1'2026
TN-201	TENAYA THERAPEUTICS	MYBPC3 associated HCM	Enrollment of additional patients is expected to continue throughout 2026 in t6E13 vg/kg MyPEAK-1 expansion cohort; Interim results from Cohort 2 & updates from Cohort 1 of MyPEAK-1 are anticipated in H1'2026; one-year Cohort 2 results & two-year Cohort 1 results are expected in H2'2026; regulatory alignment for pivotal plans also anticipated in H2'2026
TN-401	TENAYA THERAPEUTICS	PKP2 associated ARVC	Enrollment of patients in 6E13 vg/kg RIDGE-1 expansion cohort is expected to continue throughout 2026; Cohort 2 interim data expected in H2'2026; regulatory alignment on pivotal plans anticipated by YE'2026
DB-OTO	REGENERON	Genetic hearing loss	US FDA decision on the submitted BLA anticipated in H1'2026
DTX401 (Pariglasgene brecaparvovec)	ultragenyx	GSDIa	PDUFA target action date is set for Aug 23, 2026
UX701	ultragenyx	Wilson disease	Pivotal Cyprus2+ study (Phase 1/2/3) fourth cohort data anticipated in 2026
UX111	ultragenyx	Sanfilippo syndrome type A (MPS IIIA)	Expected PDUFA target action date is Sep 19, 2026
INO-3107	INOVIO	RRP	FDA assigned standard 10-month review with a PDUFA target action date of Oct 30, 2026
OCU410 (AAV5-RORA)	ocugen	Geographic atrophy secondary; dAMD	Phase 3 registrational trial expected to begin in Q3'2026
OCU400	ocugen	RP	Rolling BLA is expected to be filed in Q3 2026
OCU410ST	ocugen	ST	Phase 2/3 GARDian3 interim data expected in Q3 2026

Gene Therapy 2026: What's on the horizon? (2/2)

Asset	Company	Indication	Upcoming Milestones
Cretostimogene	CG ONCOLOGY	Intermediate-risk NMIBC	Topline Phase 3 data from PIVOT-006 trial expected in 1H of 2026
Cretostimogene+Gemcitabine	CG ONCOLOGY	High-risk NMIBC	First results from CORE-008 Cohort CX Phase 2 study expected in the 1H of 2026
Cretostimogene	CG ONCOLOGY	High-Risk BCG unresponsive NMIBC with CIS ±Ta/T1	BLA submission is expected to be completed in 2026
Cretostimogene	CG ONCOLOGY	High-Risk BCG unresponsive NMIBC without CIS with Ta/T1; High-Risk BCG naïve NMIBC with CIS ±Ta/T1	Durability data from BOND-003 Cohorts C & P and CORE-008 Cohort A expected in 2026
RGX-202	REGENXBIO	DMD	Phase 1/2 data topline pivotal results expected in early Q2 2026; FDA/EMA additional regulatory interactions planned for 1H 2026 & pre-BLA FDA meeting targeted mid-2026
Surabgene lomparvovec (sura-vec/ABBV-RGX-314)	REGENXBIO	wAMD; Diabetic retinopathy	Wet AMD: Topline data with AbbVie from ATMOSPHERE (Phase 2/3) & ASCENT (Phase 3) pivotal trials expected in Q4 2026, Diabetic retinopathy: first patient dosing in pivotal Phase 2b/3 trial is expected in Q2 2026
Isaralgagene civaparvovec (ST-920)	Sangamo THERAPEUTICS	Fabry disease	Rolling BLA submission completion expected in 2026
MCO-010	NANOSCOPE THERAPEUTICS	Stargardt disease; Geographic atrophy	Plan to initiate Phase 3 trial for Stargardt disease in 2026, alongside planned initiation of Phase 2 Geographic atrophy program in 2026
SRP-1005 (ARO-HTT)	SAREPTA THERAPEUTICS	Huntington's disease	Expected to initiate the first-in-human clinical trial (INSIGHTT) in Q2 2026
Zilganersen	IONIS	AxD	Expected PDUFA target action date is Sep 22, 2026
cemacabtagene ansegedleucel (cema-cel)	Allogene THERAPEUTICS	LBCL	Phase 2 ALPHA3 Trial expand to South Korea & Australia with patient screening & enrollment expected to begin in Q2 2026



Regulatory & Approval Landscape

(Tracking approvals & regulatory trends across the gene therapy)

From 2021 to 2025: CGT Approval Journey

2021

- **Abecma** (idecabtagene vicleucel) — 2021 — Bristol Myers Squibb — MM
- **Breyanzi** — 2021 (US) 2022 (EU) — Bristol Myers Squibb — LBCL; CLL; FL; MCL; MZL

2022

- **ADSTILADRIN** (nadofaragene firadenovec-vncg) — 2022 — FERRING PHARMACEUTICALS — NMIBC
- **CARVYKTI** (ciltacabtagene autoleucel) — 2022 (US) 2022 (EU) — Johnson&Johnson — MM
- **HEMGENIX** (etranacogene dezaparvovec-drlb) — 2022 (US) 2023 (EU) — CSL — Hemophilia B
- **Kebilidi** (eladocagene exuparvovec) / **Upstaza** (eladocagene exuparvovec) — 2022 (EU) 2023 (US) — PTC THERAPEUTICS — AADC
- **ROCTAVIAN** (valoctocagene roxaparvovec-rvox) — 2022 (EU) 2023 (US) — BIOMARIN — Hemophilia A
- **skysona** (elivaldogene autotemcel) — 2022 — bluebirdbio — CALD
- **zynteglo** (betibeglogene autotemcel) — 2022 — bluebirdbio — β-thalassemia

2023

- **casgevy** (exagamglogene autotemcel) — 2023 (US) 2024 (EU) — VERTEX — SCD; TDT
- **Elevidys** delandistrogene moxeparvovec-rokl — 2023 — SAREPTA THERAPEUTICS — DMD
- **lyfgenia** (lovotibeglogene autotemcel) — 2023 — bluebirdbio — SCD
- **Vyjuvek** beremagene geperpavec-svdt — 2023 (US) 2024 (EU) — Krystal — DEB

2024

- **Tecelra** afamitresgene autoleucel — 2024 — Adaptimmune — Synovial carcinoma
- **lenmeldy** / **libmeldy** — 2024 (US) 2025 (EU) — Orchard therapeutics — PSLI; PSEJ; ESEJ; MLD
- **BEQVEZ** fidanacogene elaparvovec-dzkt — 2024 — Pfizer — Hemophilia B
- **AUCATZYL** obecabtagene autoleucel — 2024 (US) 2025 (EU) — Autolus — ALL

>2025

- **zevaskyn** (prademagene zamikeracel) — 2025 — Abeona THERAPEUTICS — RDEB
- **WASKYRA** — 2025 (US) 2026 (EU) — FONDAZIONE Telethon — WAS
- **Papzimeos** zopapogene imadenovec-drba — 2025 — PRECIGEN — RRP
- **Otarmeni** (lunsotogene parvec-cwha) — 2026 — REGENERON — SNHL
- **itvisma** (onasemnogene abeparvovec-brve) suspension for intrathecal injection — 2025 — NOVARTIS — SMA
- **ENCELTO** (revakinagene taroretcel-lwey) — 2025 — neurotech PHARMACEUTICALS — MacTel

INSIGHTS CORNER

Note: Approved products are mapped on the timeline according to their initial approval in the US or EU

US EU

Intravenous Intravesical Intraputaminal Topical Subcutaneous Intrathecal Intravitreal Intracochlear

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

2026 Regulatory Outlook: CGT

How updates from US FDA, EMA & ICH are redefining cell & gene therapy?

- **Feb, 2026: From Plausible Mechanism to Personalized Genetic Treatment** - The purpose is to explain how individualized therapies can demonstrate safety & effectiveness based on scientifically reasonable (plausible) mechanism. It provides recommendations to help developers collect sufficient clinical safety & efficacy data. Also, ensures that the therapy is manufactured with proper quality standards & is suitable for its intended use
- **Jan, 2026: From Lab to Market Faster: Flexible CMC Framework for CGT:** US FDA is using a more flexible approach to CMC requirements for CGT, like clinical development, commercial specifications & process validation flexibilities, to help companies better prepare for BLA submission & approval
- **Sep, 2025: Fast-Track Regulatory Programs for Regenerative Medicine:** This guidance helps developers of regenerative medicine therapies for serious or life-threatening conditions understand the US FDA expedited programs, including RMAT designation. It explains how to use accelerated approval pathways, provides advice on clinical development & highlights opportunities to engage with CBER to speed up the development & review process
- **Sep, 2025: Beyond Approval: Capturing Real-World Data for CGT Products:** The aim is to explain methods for collecting safety and efficacy data for CGT products after approval, including LFTU (e.g., 15 years). (This guidance does not cover data collected to support the expansion of clinical indications)
- **Sep, 2025: Innovative Trial Designs for CGT in Small & Rare Populations:** US FDA provide recommendations to sponsors planning clinical trials for CGT products used in small or rare patient populations. It explains FDA requirements and outlines different trial designs (Adaptive/Bayesian trial design & Master protocols) and surrogate endpoints that can be used to generate the clinical evidence needed for product approval. (This document elaborates on the US FDA previously established guidance principles)

FDA U.S. FOOD & DRUG ADMINISTRATION

- **Dec, 2025: Future Guideline Roadmap for ATMPs:** ICH CGTDG, first established key guiding principles for ATMPs & then created a roadmap for CGT, focusing on priority areas that need better harmonization along with recommendations for new or updated guidelines
- **May, 2025: ATMP Comparability: Proposed New ICH Q5E Annex:** ICH Multidisciplinary guideline (ICH Q5E) is proposed to address the unique development & regulatory challenges of ATMPs, including CGTs, with a focus on ensuring comparability when their manufacturing processes changes

ICH harmonisation for better health

- **Jan, 2025: EMA Sets Standards for ATMP Clinical Trials (Effective July 1, 2025):** EMA has introduced a multidisciplinary guideline outlining the development, manufacturing, quality control, non-clinical & clinical requirements for investigational ATMPs. It explains what data and structure are needed for a CTA (both early and late-stage studies) helping companies prepare for a MAA
- **Mar, 2025: EMA Revises GMP Framework for ATMPs:** EMA has released a concept paper proposing updates to Part IV of the EU GMP guidelines for ATMPs, aiming to align with current Annex 1, while maintaining flexibility in ATMP production

EUROPEAN MEDICINES AGENCY
SCIENCE MEDICINES HEALTH

INSIGHTS CORNER



Case Study: Tracking Spark Therapeutics Growth Within Gene Therapy Ecosystem

(Corporate evolution, strategic shifts & ecosystem development)

The Spark Therapeutics Journey

From Academic Labs to Gene Therapy Pioneer

Spark Therapeutics under Roche:

Roche's Spark acquisition serves as a cautionary tale in gene therapy commercialization, with most of its once-promising pipeline discontinued amid LUXTURNA's declining sales and Hemophilia setbacks, prompting a full strategic reset

The sole surviving asset, high-risk/high-reward RG6662 for Huntington's disease now entering P1/2, will prove pivotal over the next 2-3 years in salvaging Roche's gene therapy ambitions

Marketed product	
LUXTURNA (Voretigene Neparvovec-rzyl)	RPE65 mutation-associated retinal dystrophy
BEQVEZ (Fidanacogene Elaparvovec)	Hemophilia B
Active Clinical Pipeline	
RG6662 (SPK-10001)	Huntington's disease
Discontinued	
SPK-8016	Hemophilia A: Axed in 2023,removed from mid-phase pipeline
SPK-3006	Pompe disease discontinued 2023, due to strategic portfolio prioritization
SPK-8011	Hemophilia A: Discontinued Dec 2024, P3 trial withdrawn before enrolling patients
SPK-7001	Choroideremia (inherited blindness) discontinued in 2025

Pre-2013

Origin: Spark's roots are in two decades of academic AAV gene therapy work at **Children's Hospital of Philadelphia (CHOP)** and the **University of Pennsylvania**

2013

Mar 2013: CHOP formally spun out this program as Spark Therapeutics. Early pipeline centered on AAV-mediated gene transfer for Hemophilia, RPE65-mediated retinal dystrophy (later LUXTURNA), and other inherited retinal and Lysosomal storage diseases

2014

Dec 2014: Global collaboration with **Pfizer** for SPK-FIX program in Hemophilia B

2015

Jan 2015: Spark became a public company via a $161M IPO trading under the ticker $ONCE signaling Spark's core thesis: one-time curative gene therapies

2017

Dec 2017: Historic **FDA Approval of First Gene Therapy: LUXTURNA** for biallelic RPE65 mutation-associated retinal dystrophy

2018

Jan 2018: Agreement with **Novartis** to develop and commercialize Luxturna outside the US
Spark announced two novel payer programs: contracting model with an option for direct-to-payer contracting, and an outcomes-based rebate arrangement

Jul 2018: : Pfizer initiated a Phase 3 program for the Hemophilia B gene therapy program (SPK-9001)

By end of **June 2018, 81% of commercial lines had "favorable" coverage of LUXTURNA**

2019

The Roche Acquisition: $4.3 Billion Exit
Completed in Dec 2019, Spark continued to operate as an **independent company** within the Roche Group.
Strategy behind: Roche wanted to secure its foothold in gene therapy and hemophilia, while Spark gained the scale and distribution muscle of a global pharma giant

2023

Spark also expanded into neurodegenerative diseases, with Roche supporting clinical work on Huntington's disease gene therapy candidate

2023

SPK-8011 for Hemophilia A entered P3 clinical trials; SPK-8016 for Hemophilia A and SPK-3006 for Pompe disease discontinued

2025

Reorganization followed a strategic review of Spark's prospects in H2 2024 leading to **Spark being fully integrated into the Roche Group**

2024

Fidanacogene elaparvovec (SPK-9001) for Hemophilia B developed with Pfizer, received FDA approval in 2024; SPK-8011 discontinued due to strategic decision

INSIGHTS CORNER

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com



Challenges, Risks & Strategic Assessment

(Limitations, barriers, opportunities & future readiness)

INSIGHTS CORNER

Gene Therapy's Biggest Hurdles



1. Safe and precise delivery to the right tissue - Efficient, tissue-specific delivery of DNA/RNA or gene-editing machinery remains one of the biggest barriers, especially for organs like the brain and heart. Systemically administered vectors must survive degradation, cross biological barriers (e.g., blood–brain barrier), and avoid off-target transduction, which current AAV and nanoparticle systems still do imperfectly



2. Immunogenicity and limited re-dosing - Viral vectors such as AAV can trigger innate and adaptive immune responses against capsid and/or transgene, which may cause toxicity and can block repeat dosing. Pre-existing neutralizing antibodies and treatment-induced immunity often exclude patients from trials or prevent them from receiving a second dose if the first one is sub-therapeutic



3. Off-target effects and long-term safety - CRISPR and related editors can introduce unintended genomic changes, raising concerns about genotoxicity, oncogenesis, and other adverse effects that may only emerge years later. Reviews emphasize that, despite improved guide design and high-fidelity nucleases, fully characterizing off-target edits in vivo and predicting their long-term consequences remains an unresolved challenge



4. Manufacturing scalability, cost, and quality control - Current manufacturing processes for viral vectors and complex delivery systems are expensive, capacity-limited, and technically demanding, which constrains patient access and slows development. 2025-2026 analyses highlight persistent pain points around large-scale GMP production, assay comparability, and CMC expectations, all of which add significant cost and time to each program



5. Clinical, regulatory, and access complexity - Gene therapy programs often face difficult trial design and endpoint selection, especially in rare diseases, alongside extensive long-term safety follow-up requirements. Even after approval, adoption can remain limited due to complex regulatory expectations, specialized treatment-center requirements, high costs, and reimbursement uncertainty

INSIGHTS CORNER

Gene Therapy: SWOT Analysis

S STRENGTHS

- **Convenience:** Potentially curative, one-time treatments for monogenic and select acquired diseases, reducing long-term disease burden
- **Rare disease success:** Demonstrated efficacy of AAV-based gene replacement in rare disorders, e.g. ZOLGENSMA, a one-time SMN1-targeted therapy for infants with SMA
- **Expanding disease reach:** Gene therapy is moving beyond rare monogenic disease into oncology, monogenic disorders (e.g. SMA), polygenic conditions (e.g. Autism), neurodegenerative diseases (e.g. Parkinson's) & infectious diseases like HIV

W WEAKNESSES

- **High cost:** Gene therapies carry high upfront prices to recover R&D from one-time dosing (e.g. GLYBERA ~€1M), leading to reimbursement barriers and market withdrawal
- **Immune response challenge:** Viral vector gene therapies (e.g. AAV) trigger CD8+ T-cell responses and inflammation, often requiring steroid-based immunosuppression across hepatic, systemic and ocular indications
- **Long-term durability uncertainty & limited RWE:** Gene therapies face LTFU challenges, with limited real-world data to assess long-term efficacy and late safety risks (e.g. oncogenesis)

O OPPORTUNITIES

- **Market size:** Growing R&D and investment pipeline, with accelerating global approvals projecting double-digit market CAGR through the next decade
- **Rare genetic diseases:** ~95% of rare genetic diseases lack approved treatments, representing significant unmet need addressable through gene therapy targeting root genetic defects
- **Advancing non-viral delivery (LNPs):** Lipid nanoparticles enable safer, scalable and less immunogenic mRNA/siRNA delivery, supporting approved therapies and enabling repeat dosing and broader *in vivo* applications

T THREATS

- **Reimbursement failures:** Payment-model failures can derail gene therapies despite regulatory approval (e.g. GLYBERA withdrawn in EU due to lack of demand, not safety)
- **Regulatory bodies demanding real-world outcomes data:** Increasing demand for robust real-world and long-term outcomes data delays approvals, constrains reimbursement & adds pricing uncertainty for gene therapies
- **Competition:** Competitive pressure from other advanced modalities (e.g. RNA therapies, small molecule or biologic functional cures)

INSIGHTS CORNER

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com



Future Frontier of Gene Therapy

(Long term outlook, innovation trends and future potential)

Gene Therapy: Future Directions



1. Shift toward *in vivo* gene editing

Gene therapy is moving from *ex vivo* edited cells toward *in vivo* delivery of genome editors directly to target tissues, aiming for simpler, one-time treatments. A 2025 review highlights rapid translation of CRISPR, base, and prime editors into *in vivo* applications across genetic diseases, beyond earlier *ex vivo* only approaches.

2. Precision targeting & safer editing platforms

Precision targeting and safer editing platforms - The field is moving toward next-generation editors and delivery systems that can selectively reach target tissues, reduce off-target effects, and avoid the risks associated with less controlled genome modification. Base editing, prime editing, and engineered delivery vehicles are expected to play a central role in making gene therapy safer and more precise.

3. Non-viral delivery

Next-generation non-viral carriers (lipid nanoparticles, exosomes/extracellular vesicles, virus-like particles, and protein-based carriers) are emerging to reduce immunogenicity and enable repeat dosing. A 2026 open-access review describes a coming "age of CRISPR delivery with non-viral biologics," emphasizing engineered exosomes and other biologic nanoparticles as key to safer, more flexible *in vivo* editing.

4. Integration of AI & data-driven development

AI and machine learning are increasingly used to design vectors and guides, predict off-target risk, optimize dose, and streamline clinical-trial operations, and are expected to be deeply embedded in future gene-therapy R&D. 2026 life-sciences forecasts describe advanced AI as the "engine" of next-gen cell and gene therapy, impacting discovery, trial design, and real-world evidence generation for regulatory and access decisions.

5. Broader indications, access, & equity focus

Gene therapy is projected to move from rare monogenic diseases into more common indications (neuromuscular, autoimmune, cardiovascular, and metabolic diseases) while simultaneously tackling access barriers. Policy and market-access trend reports from 2024–2026 highlight outcome-based payment models, community-site delivery, and global trial diversification as crucial to making these high-value therapies more widely available.

INSIGHTS CORNER

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com



Glossary of Abbreviations

(Key terms used in gene therapy)

Abbreviations (1/2)

Abbreviations	Full Form
AADC	Aromatic L-amino Acid Decarboxylase
AATD	Alpha-1 Antitrypsin Deficiency
AAV	Adeno-associated Virus
ALL	Acute Lymphoblastic Leukemia
ARVC	Arrhythmogenic Right Ventricular Cardiomyopathy
ATMPS	Advanced Therapy Medicinal Products
AXD	Alexander Disease
BLA	Biologics License Application
CAGR	Compound Annual Growth Rate
CALD	Cerebral Adrenoleukodystrophy
CBER	Center For Biologics Evaluation and Research
CGT	Cell and Gene Therapy
CIS	Carcinoma *In Situ*
CLL	Chronic Lymphocytic Leukemia
CMC	Chemistry, Manufacturing and Controls

Abbreviations	Full Form
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeat
CTA	Clinical Trial Application
DAMD	Dry Age-related Macular Degeneration
DEB	Dystrophic Epidermolysis Bullosa
DMD	Duchenne Muscular Dystrophy
ESEJ	Early Symptomatic Early Juvenile
FL	Follicular Lymphoma
GSDIA	Glycogen Storage Disease Type Ia
HCM	Hypertrophic Cardiomyopathy
HIV	Human Immunodeficiency Virus
HTA	Health Technology Assessment
IPO	Initial Public Offering
LBCL	Large B-cell Lymphoma
LNPS	Lipid Nanoparticles
LTFU	Long Term Follow-up

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

Abbreviations (2/2)

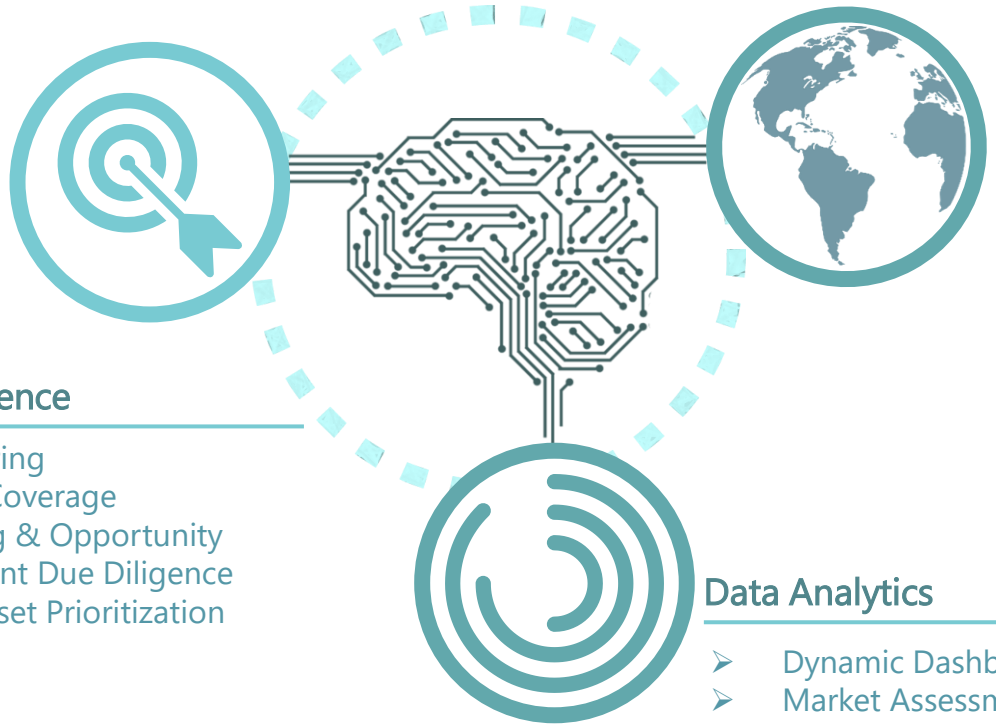
Abbreviations	Full Form
MACTEL	Macular Telangiectasia
MCL	Mantle Cell Lymphoma
MLD	Metachromatic Leukodystrophy
MM	Multiple Myeloma
MRNA	Messenger RNA
MYBPC3	Myosin Binding Protein C3
MZL	Marginal Zone Lymphoma
NMIBC	Non-muscle Invasive Bladder Cancer
PKP2	Plakophilin-2
PSEJ	Pre-symptomatic Early Juvenile
PSLI	Pre-symptomatic Late Infantile
RDEB	Recessive Dystrophic Epidermolysis Bullosa
RMAT	Regenerative Medicine Advanced Therapy
RP	Retinitis Pigmentosa
RRP	Recurrent Respiratory Papillomatosis

Abbreviations	Full Form
RRP	Recurrent Respiratory Papillomatosis
RWE	Real World Evidence
SCD	Sickle Cell Disease
SCID	Severe Combined Immunodeficiency
SHTG	Severe Hypertriglyceridemia
SIRNA	Small Interfering RNA
SLL	Small Lymphocytic Lymphoma
SMA	Spinal Muscular Atrophy
SMN1	Survival Of Motor Neuron 1
SNHL	Sensorineural Hearing Loss
ST	Stargardt Disease
TDT	Transfusion-dependent B-thalassemia
TKA	Total Knee Arthroplasty
WAMD	Wet Age-related Macular Degeneration

This presentation is non-exhaustive. For additional details or clarifications, please feel free to contact us at bd@octavusconsulting.com

ABOUT COMPANY

Octavus delivers evidence-based consulting services to empower our clients in making well-informed decisions.



Market Research

- Quantitative & Qualitative Research
- Awareness, Trial & Usage (ATU) Studies
- KOL & PAG Identification & Mapping
- Concept/Product Testing
- AI & Human Confluence Transcription
- Content Analysis & Translation

Business Intelligence

- CI & Monitoring
- Conference Coverage
- Market Sizing & Opportunity
- Pre-investment Due Diligence
- Company/Asset Prioritization

Data Analytics

- Dynamic Dashboarding & Reporting
- Market Assessment & Epidemiology Analysis
- Sales Reporting & In-line Forecasting
- Social Media Analytics
- Clinical Analytics



OCTAVUS CONSULTING
Data.Insights.Decisions.

Contact Us



Himanshu Sehgal
himanshu@octavusconsulting.com
Managing Partner



Rakesh Singh
rakesh@octavusconsulting.com
BD Manager

